

Response to Reviewers

Paper: "Green Browsing – Comparing Energy Consumption of Modern Web Browsers"

We thank all three reviewers for their careful reading of the paper and their constructive comments. We have revised the paper accordingly. Below we address each comment point by point; reviewer comments are quoted in italics, followed by our response and, where applicable, a short description of the corresponding change made to the paper. Line/section references refer to the revised paper.

Reviewer 1 (Weak Accept)

Major Issues

Comment 1

"The measurement pipeline is only described at a very high level. Key implementation details (e.g., power-meter model, sampling rate, calibration procedure, synchronization method between power and performance counters, analysis scripts etc.) are missing, limiting true reproducibility."

We agree and have substantially expanded the description of the measurement pipeline in Section 2.4. We now name the power meter used (GUDE Expert Power Control 1202), specify the sampling frequency (1 Hz), and state that the device was used in its factory-calibrated state. We clarify that only one power-metering device was used for the analysis presented, because the focus of this study is the energy consumption of the Measurement Client. We also added an explicit description of the offline synchronization procedure: electrical power measurements, Windows Performance Monitor logs, and Actiona event logs were aligned using timestamps, with system clocks synchronized via an NTP server prior to each run. Finally, we clarify that timestamp alignment, baseline correction, aggregation, statistical analysis, and visualization were all performed using custom Python scripts.

Comment 2

"The scenario automation with Actiona ensures repeatability, but only if the Actiona scenarios are also provided in some form, which I did not find in the replication package."

The Actiona automation scenarios for all browsers have now been added to the Zenodo replication package and are linked from the paper: Baseline (10.5281/zenodo.21081950), Brave (10.5281/zenodo.21139042), Chrome (10.5281/zenodo.21082941), Edge (10.5281/zenodo.21139455), Firefox (10.5281/zenodo.21138748), and Opera (10.5281/zenodo.21139326).

Comment 3

"Since separate measurements of client and server systems were conducted, the server's power consumption is measured but never reported or contextualized with the client's figures. [...] I would strongly suggest clarifying the server measurements and providing them as well."

We have expanded Section 2.4 to explain the rationale for the local server setup: it provides control over content delivery (no caching effects) and reduces network variability. We clarify that an Internet connection was nevertheless retained so that browsers could still trigger background activity (updates, telemetry, sync). We also clarify that, while the server's power consumption was measured for monitoring purposes, it was not the object of our research questions — the study focuses exclusively on the browser (client-side) energy consumption — and it is therefore not reported as a separate result.

Comment 4

"The scenario is highly synthetic [...] real-world browsing includes heterogeneous content and variable network latency [...] I suggest the authors go into more detail on how the chosen scenario maps to typical user behavior."

We acknowledge this limitation more explicitly. In Section 2.5 we added a sentence clarifying that the standardized workload should be interpreted as a representative benchmark of common browser activities — developed in a focus group with informatics-related users aged 20–40 — rather than a complete model of everyday web usage. This limitation is also revisited in Section 4.2 (Critical Evaluation of the Results).

Comment 5

"The inclusion of DNS calls via Pi-hole adds a novel dimension [...] the analysis potentially conflates DNS request count with "background activity" without confirming the destination domains. Additionally, why is there a connection to the internet at all, if the whole scenario is run in a local network?"

We substantially expanded the network traffic analysis in Section 3.5. We attempted a categorization of destination domains from the Pi-hole logs but found this was not reliably possible with the available data (e.g., no per-call byte information), and we now state this limitation explicitly rather than drawing unsupported conclusions from domain names alone. The network metric is now explicitly described as reflecting primarily communication with external endpoints, since the test website itself is hosted locally; traffic is now also broken down into sent and received data per browser (Table 5). Regarding Internet access: we clarify that the test website is hosted locally to control content delivery, while browsers retain Internet access specifically so that background communication (telemetry, updates, sync) — a relevant aspect of real-world browser behavior — remains observable via Pi-hole.

Comment 6

"Section 3.5 states "Brave [...] blocks many third-party tracking and telemetry domains while concentrating its remaining communication on few hosts." [...] Why does Brave generate more traffic and why do you conclude that this is somehow connected to it being more privacy focused?"

We agree this causal claim was not substantiated by our data. This statement has been removed from Section 3.5. We now report the divergence between traffic volume and DNS request count purely descriptively, without attributing it to a specific cause.

Comment 7

"Absence of statistical significance testing weakens confidence in conclusions."

We agree that no formal significance testing (e.g., paired t-tests or Wilcoxon signed-rank tests) was performed, and we now state this explicitly as a limitation in Section 4.2, leaving formal significance testing for future work. In addition, we identified and corrected an error in how the standard deviation was originally computed: it had been calculated across all individual measurement points within the 20 runs rather than across the 20 run-level means, which inflated the reported SD values (e.g., Chrome's SD of 11 W noted by the reviewer). The SD has been recalculated based on the mean values of the 20 individual runs and the corrected values are now reported in Tables 1–6.

Comment 8

"Where does Tiny11 originate? I strongly suggest adding a reference or link. [related to hardware/OS scope]"

A reference to the Tiny11 project (NTDEV, GitHub) has been added, together with a short description of Tiny11 and how it differs from a standard Windows 11 installation (removal of preinstalled productivity, multimedia, gaming, communication, and cloud-integration components) in Section 2.4.

Minor Issues

Comment	Response / Action Taken
Author list formatting on page 1 does not follow the template	Corrected: author names, affiliations, and formatting now follow the conference template.
Figure text (measurement setup, time-series plots) too small to read in print	Figures were regenerated at higher resolution with larger axis/legend labels.

Comment	Response / Action Taken
Figure 2: add looping conditions	The loop condition ("20 repetitions reached?") is now shown explicitly in the flowchart.
Figure 3: "calles" → "calls"; inconsistent units (RAM has a unit, other metrics do not)	Spelling corrected; units clarified consistently across the figure.
Inconsistent citation style (mixing English/German, missing access dates, capitalization)	Citation style has been unified to English throughout, and access dates were added where missing.
Origin of Tiny11 unclear, no reference	Reference and short description added (see Major Issue 8 above).
Inconsistent use of "baseline" vs. "idle" terminology (Section 2.5 vs. 2.6)	Terminology has been unified throughout the paper: "Idle Base measurement" is now used consistently, with clear definitions in Section 2.6.
SD in Table 1 appears very high (>50% for Chrome); is it representative?	As noted under Major Issue 7, this was due to an error in the SD calculation (computed across all measurement points rather than per-run means). This has been corrected; the recalculated SD values are now much smaller and consistent with the plotted time series.
What does "Average" mean in Figures 4–7?	"Average" now explicitly refers to the mean power/CPU/RAM/disk value across the 20 runs at each time step; this is defined in the figure captions and in Section 2.6.

Reviewer 2 (Weak Accept)

Major Concern

Comment 1 (Novelty)

"My concern with the paper is the lacking novelty. I remember papers on this subject since more than 15 years [...] please check e.g. previous EnviroInfo proceedings. [...] A major problem previous approaches had was to restrict the measured power consumption only to the measured scenario [...] the Measurement Setup Section does not give details on how this was solved."

We agree that browser energy studies have a long history, including within the EnviroInfo community, and we have revised the Introduction (Section 1) to position our contribution more clearly against this prior work. We now explicitly discuss Dick et al. (EnviroInfo 2011, 2012), which introduced standardized browser benchmark scenarios with four browsers (Chrome, Firefox, Opera, IE) on older hardware, and Kleinert & Arndt (2023), which focused on server-side load distribution rather than client-browser comparison. We highlight what has changed since: today's browser landscape is dominated by the Chromium/Blink engine (four of five browsers tested), tracking and telemetry are far more pervasive than in earlier studies, the Blue Angel eco-label criteria were updated in 2025, and DNS-level background-activity monitoring (via Pi-hole) has received little attention in prior browser energy studies. We frame our contribution as the first synchronized measurement of power, CPU, RAM, disk, and DNS activity for modern (2025) browsers under the updated Blue Angel criteria, with openly available data. Regarding isolating scenario power from background/parallel activity: this is now described explicitly in Section 2.6. We conducted a separate Idle Base measurement (system idle, no browser or other application running) and subtracted it from both the Base and Scenario measurements on a per-timestamp basis to isolate the energy attributable to browser execution. In addition, Tiny11 was used specifically to minimize background services and parallel activity that could confound the scenario-level measurements, as now described in Section 2.4.

Minor Issues

Comment	Response / Action Taken
Spell out DNS at first mention	Done.
Spell out DOM at first mention	Done.

Comment	Response / Action Taken
Spell out CPU at first mention	Done.
Spell out RAM at first mention	Done.
Briefly explain Actiona	A short explanation has been added at first mention in Section 2.4 (open-source, cross-platform input-layer automation tool).
Briefly explain Pi-hole	A short explanation has been added at first mention in Section 2.4 (open-source, network-level DNS resolver, used here as a passive DNS monitoring component).

Reviewer 3 (Strong Accept)

We thank the reviewer for the positive assessment of the paper's contribution to the EnviroInfo conference and to Green Software Engineering. All requested clarifications have been addressed.

Comments

Comment	Response / Action Taken
Briefly explain the software tool Actiona	Addressed — see Reviewer 2, Minor Issues.
Write out the abbreviation DNS when introduced	Addressed — see Reviewer 2, Minor Issues.
Briefly explain the tracking and ad-blocking tool Pi-hole	Addressed — see Reviewer 2, Minor Issues.
Write out the abbreviation DOM when introduced	Addressed — see Reviewer 2, Minor Issues.
Write out the abbreviation CPU when introduced	Addressed — see Reviewer 2, Minor Issues.
Write out the abbreviation RAM when introduced	Addressed — see Reviewer 2, Minor Issues.

Summary of Changes

In summary, the revised paper includes: (1) a substantially expanded and more detailed description of the measurement setup (power meter, sampling rate, synchronization, analysis pipeline); (2) publication of the Actiona automation scenarios in the Zenodo replication package; (3) an expanded and corrected network/DNS traffic analysis, including removal of an unsubstantiated claim regarding Brave; (4) correction of an error in the standard-deviation calculation, now computed across per-run means; (5) an expanded discussion of novelty relative to prior EnviroInfo work and a clearer description of how scenario power was isolated from background/idle activity; (6) a description and reference for Tiny11; (7) consistent terminology (baseline vs. idle), consistent units, corrected figures, and unified citation style; and (8) all requested abbreviations and tools (DNS, DOM, CPU, RAM, Actiona, Pi-hole) explained or spelled out at first mention. We believe these revisions have substantially strengthened the paper and thank the reviewers again for their valuable feedback.

For the creation of this response to reviewers, we used Claude to help finalize the response.